Examiner only

Answer **all** questions.

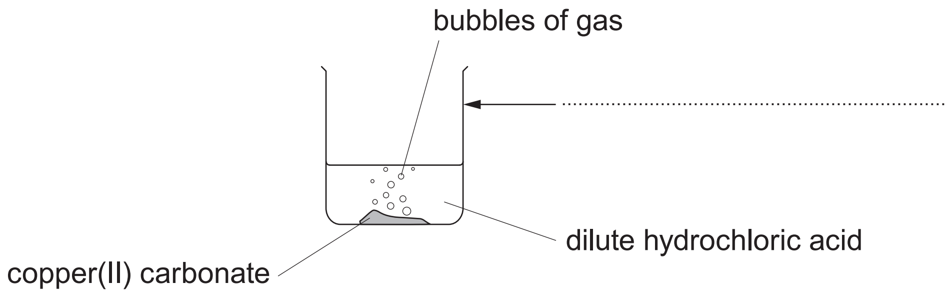
1. Selwyn carried out an experiment to prepare copper(II) chloride crystals.

The diagrams show the stages of the experiment he carried out.

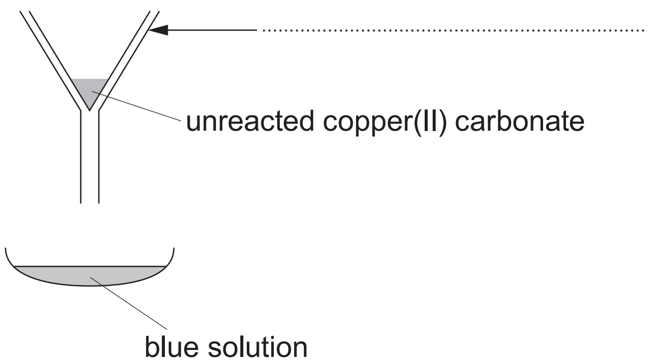
(a) Choose apparatus from the box to label the diagrams. [4]

evaporating basin	**conical flask**	**filter funnel**	**electronic balance**
filter paper	**beaker**	**test tube**	

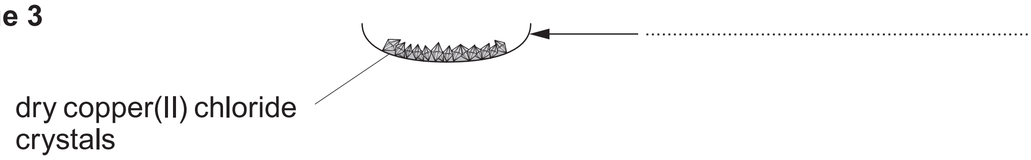
Stage 1



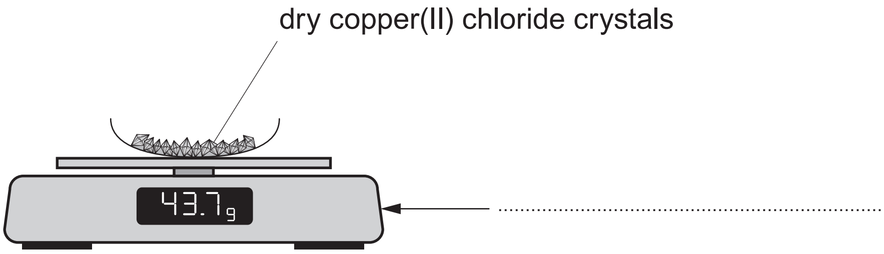
Stage 2



Stage 3



Stage 4



Examiner only

(b) In **Stage 1** Selwyn added copper(II) carbonate until **all** the dilute hydrochloric acid was used up.

Tick (✓) the box next to the statement which best describes what Selwyn would see when all the acid had been used up. [1]

bubbling increases	☐
bubbling stops	☐
bubbling decreases	☐

(c) The gas formed in **Stage 1** turns limewater milky. <u>Underline</u> the name of this gas. [1]

oxygen **hydrogen** **carbon dioxide** **nitrogen**

(d) Choose words from the box to complete the following sentences. [2]

evaporation	**filtration**	**distillation**	**neutralisation**

The process used to remove the unreacted copper(II) carbonate in **Stage 2** is called

....

The process used to remove water in **Stage 3** is called ....

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(e) The container holding the crystals has a mass of 29.8 g. Using the information given in **Stage 4**, calculate the mass of the crystals formed. [1]

Mass of crystals formed = ... g

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